

SRE Project Proposal Submission Form

DO NOT SUBMIT HANDWRITTEN FORM

SEM4/CUI/SE-WAH/26

* Project Title:

(150 characters)

AlgoVision – AI Powered Data Structures Learning & Analysis System

* Which real world problem shall be solved by this FYP?

(800 characters)

Students often struggle to understand how data structures and algorithms work internally because most learning resources focus only on theoretical explanations or static code examples. Without visualization and practical experimentation, it becomes difficult for learners to understand how operations like insertion, deletion, traversal, or sorting actually occur step-by-step. **AlgoVision** solves this problem by providing an interactive platform where students can **visualize algorithms, execute code, analyze performance, compare data structures**, and receive **AI-assisted guidance**. The system integrates algorithm visualization, code execution, performance analysis, and an AI tutor to create an intelligent learning environment for data structures education.

Detail of Project Members:

Muhammad Akif Naveed FA24-BSE-129 (Team Lead)

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Supervised By:

Dr. Wasif Nisar

Project Streams:

☒ Web-based FYPs

☐ Desktop Applications

☐ Mobile Apps FYPs

☐ Game-based FYPs

☐ Hardware-based

* Project Description:

(3000 characters)

Note: Login/logout and authentication/authorization shall be the default functionalities of Web/Gaming/Mobile/Desktop FYPs.

(a) What is the overall working/summary of this FYP? (1000 characters)

AlgoVision – AI Powered Data Structures Learning & Analysis System is an interactive platform designed to help students understand and experiment with data structures and algorithms through visualization, execution, and analysis. The system allows users to perform operations on structures such as arrays, linked lists, stacks, queues, trees, graphs, sorting, searching, and hashing while visualizing each step of the algorithm in real time. It includes a built-in code playground where users can write and execute programs to observe algorithm behavior and outputs. The system also analyzes execution performance by measuring time and memory usage and presenting results through graphical comparisons between algorithms. An integrated AI tutor provides explanations, hints, and learning assistance to help users better understand concepts and solve practice problems. The platform also stores performance results and generates downloadable reports for analysis and learning progress tracking.

(b) Write name and detail of each module in your FYP.

Modules of the System
1. User Management Module
This module manages user authentication and authorization functionalities including registration, login, logout, and profile management. It allows users to securely access the platform, store their activity history, and track their learning progress.
2. Data Structure Visualization Module
This module provides interactive visualization of different data structures including arrays, linked lists, stacks, queues, trees, graphs, and hashing structures. Users can perform operations such as insertion, deletion, searching, and traversal while observing step-by-step algorithm execution through animations.
3. Algorithm Performance Analysis Module
This module analyzes algorithm efficiency by measuring execution time and memory usage. It also generates graphical comparisons between different algorithms (e.g., sorting or searching algorithms) to help users understand performance differences and time complexity behavior.
4. Code Execution (Compiler) Module
This module provides an integrated coding playground where users can write, compile, and execute programs in supported languages such as C++ and Python. The module displays execution output and helps users experiment with algorithms directly within the system.
5. AI Tutor Support Module
This module integrates an AI-based assistant that provides explanations of data structures and algorithms, answers conceptual questions, and offers hints for solving practice problems. The AI tutor helps improve user understanding and learning efficiency.
6. Practice Problem Module
This module provides structured practice questions related to various data structures and algorithms. Users can attempt problems, test solutions using the code execution environment, and receive guidance from the AI tutor.
7. Reporting and Performance Tracking Module
This module records algorithm execution results and user performance data. It generates analytical reports including graphs of time and memory usage and allows users to export performance reports in PDF format for learning analysis.

1: Which module shall be developed by student-1?

(500 characters)

Student-1 will develop the backend and intelligent components of the system. Responsibilities include integrating the AI Tutor module using the **Groq API**, implementing the **code execution (compiler) module**, and building the **algorithm performance analysis system** for measuring time and memory usage. Student-1 will also design and manage the database and develop **backend APIs** for communication between the frontend, **execution engine**, and **database**.

2: Which module shall be developed by student-2?

(500 characters)

Student-2 will develop the **system interface** and **visualization components**. Responsibilities include designing the user interface and implementing the data structure visualization module for arrays, linked lists, stacks, queues, trees, and graphs. Student-2 will also develop the **practice problem interface** and **integrate** the frontend with backend services to enable smooth interaction between user operations, visualizations, and algorithm executions.

(d) Were similar FYPs already developed on the same topic in your department?

Yes ☐ No ☒

(d-1) Copy name(s) of latest one or two similar FYPs from RMS student console and paste below. (150 characters)

(d-2) Mention below the three (3) new, but main, functionalities you are adding to this FYP. (600 characters)

1.

2.

Development Environment:

- **Tools** (e.g.Dot Net platform, Android Studio, Xcode, Swift, Ionic, Xamarin, PhpStorm
- /Php Laravel, WordPress, Maya, Unity 3D, Photoshop, MATLAB, ns-2, Python,
- Java EE, Java ME, NetBeans, Java Script, Node.js, Angular.js, JSON, OpenCV)
- **DBMS** (e.g. SQL Server, MySQL, SQLite, Oracle, Teradata)
- **Platform** (e.g. Windows, Linux

Tool(s):
Python (FastAPI), React.js, Next.js, Tailwind CSS, D3.js, Chart.js, Monaco Editor, Groq API (AI Integration), Judge0 API (Code Execution), Git, GitHub, Visual Studio Code, Streamlit/HuggingFace Spaces for deployment.

DBMS:
PostgreSQL (via Supabase)

Platform:
Web-Based Application deployed on Cloud Platform (HuggingFace Spaces / Streamlit Cloud), Development Environment on Windows/Linux.

Evaluation Criteria for 7th Semester:

(Week 14 - 16)

1. SRDS (Functional & Non-functional requirements, Use case diagram, Sequence diagram, Class diagram, Entity relationship diagram / Detailed hardware configuration)
2. Implementation of ONE major use case ($\geq 30\%$ FYP work), which does not include login/logout.
3. Interface with complete functionality of major use case / In case of hardware, provide configuration of major use case functionality.

Evaluation Criteria for Internal 8th Viva:

(Week 12 & 13)

1. Implementation of all use cases ($\geq 90\%$ FYP work)
2. Project in running and working form as per Use case, Class and Sequence diagrams as mentioned in SRDS
3. Initial FYP report
4. Deployment
 - Web: Your website must be online on any free/paid hosting service
 - Mobile App: Your app must be in APK/iOS form so that it can be installed
 - Gaming: Your game must be in executable form so that it can be installed
 - Desktop Application: Your application must be in executable form so that it can be installed
 - Hardware: Your final product must be in proper casing and should give look and feel of sellable item